

DAY-1 — Linux Fundamentals

CyberNotes • Detailed Beginner Reference (Improved Edition)

Learning goal: Build a practical, working understanding of **linux fundamentals**, connect the concept to real cybersecurity work, and practice it safely in a local lab.

Study rule: Understand a command or concept before copying it. Security tools are powerful because they interact with real systems — your practice environment should always be intentionally controlled and authorized.

1. Core Concepts

Core idea: Linux Fundamentals is a building block for practical cybersecurity. This section explains what each term actually means, where you'll encounter it, and how it connects to the bigger picture.

Why it matters: Security work depends on understanding normal system and network behavior before you can recognize abnormal behavior. This topic connects directly to Linux, networking, web security, monitoring, and troubleshooting — the threads running through this whole 20-day series.

Vocabulary covered: Linux, distributions, filesystem, terminal, paths, files, directories, basic commands, permissions.

Reference	What to remember
<code>pwd</code>	show current directory
<code>ls -la</code>	list files including hidden files
<code>cd <dir></code>	change directory
<code>mkdir <dir></code>	create directory
<code>touch file</code>	create file
<code>cp A B</code>	copy
<code>mv A B</code>	move/rename
<code>rm file</code>	remove file carefully
<code>cat file</code>	display text
<code>chmod</code>	change permissions

2. Concept Explanations

Each term below is explained in plain language, with a note on why it matters for security work. Read these before running any commands.

1. Linux

Linux is an open-source operating system kernel that powers most servers, cloud infrastructure, and security tools. Unlike Windows, its design is transparent and text-driven, which is why almost every professional security tool (nmap, Wireshark, Burp Suite, Metasploit) runs natively on it. Learning Linux is the foundation for everything else in this course.

2. distributions

A distribution ("distro") is a specific packaging of the Linux kernel plus software, package manager, and defaults — examples include Ubuntu, Debian, Kali, and CentOS. Security professionals commonly use Kali or Parrot OS because they ship pre-loaded with pentesting tools. Knowing which distro you're on matters because commands and package managers differ (apt vs yum vs pacman).

3. filesystem

The Linux filesystem is a single tree rooted at `/`, with standard directories like `/etc` (configuration), `/var/log` (logs), `/home` (user files), and `/bin` (programs). Everything — including devices and processes — is represented as a file somewhere in this tree. Understanding this layout tells you where to look for evidence during an investigation.

4. terminal

The terminal is a text interface to the shell, where you type commands instead of clicking icons. It gives precise, scriptable control over the system and is how virtually all security tooling is operated. Comfort with the terminal is what separates someone who can follow a tutorial from someone who can actually investigate a system.

5. paths

A path describes the location of a file or directory in the filesystem tree. Absolute paths start from `/` (e.g. `/home/user/notes.txt`) and always point to the same place regardless of where you are; relative paths (e.g. `./notes.txt` or `../notes.txt`) depend on your current directory. Confusing the two is a very common source of "file not found" errors.

6. files

A file is a named unit of stored data — text, a program, a config, or a log. In Linux, file type isn't determined by its extension (unlike Windows); the actual content and permission bits decide what can be done with it. Commands like `file` and `cat` help you confirm what you're actually looking at before acting on it.

7. directories

A directory (folder) organizes files and other directories into a hierarchy. Every directory contains at least two special entries: `.` (itself) and `..` (its parent), which is how `cd ..` moves you up a level. Clear directory structure — like keeping each day's lab in its own folder — makes your work reproducible and easy to audit later.

8. basic commands

Basic commands (`pwd`, `ls`, `cd`, `mkdir`, `touch`, `cp`, `mv`, `rm`, `cat`) are the small set of tools you use constantly to navigate and manipulate the filesystem. Each does one job well, and they're commonly chained together to build more complex workflows. Mastering these first makes every later topic — networking, scripting, log analysis — much faster to learn.

9. permissions

Permissions (read/write/execute, for owner/group/others) control who can do what with a file, and `chmod` is the command that changes them. Getting permissions wrong is one of the most common real-world causes of both broken systems and security vulnerabilities — a file that's too open can leak data, one that's too restrictive breaks the application. This links directly into Day 4, where permissions get a full deep dive.

3. Safe Practical Lab

Step 1. Define the lab target and the exact question you are trying to answer.

Step 2. Run the smallest safe observation or command that can answer that question.

Step 3. Read the output carefully; record important fields, errors, timestamps, and assumptions.

Step 4. Change one variable at a time and compare the result.

Step 5. Write a short conclusion and clean up temporary files or services.

Example lab sequence:

```
pwd
mkdir -p ~/cybernotes/day1
cd ~/cybernotes/day1
touch notes.txt
ls -la
```

4. Common Mistakes & Troubleshooting

- Running a command without checking its target, current directory, or privileges.
- Confusing a tool's output with proof of a vulnerability or compromise — output is evidence, not a conclusion.
- Ignoring error messages instead of reading the exact reason for failure.
- Changing several settings at once, making it hard to tell what actually fixed or broke the lab.
- Saving real credentials, tokens, or personal data in practice notes or public repositories.
- Testing a public target just because it's reachable — reachability is never the same as authorization.

Troubleshooting checklist:

- What system am I on?
- What exact command/request did I run?
- What did I expect?
- What happened instead?
- What does the error message say?
- Can I reproduce it in a clean local lab?
- What changed since the last successful attempt?

5. Deep-Dive Questions

Answer these in your own words after finishing the lab — they're designed to check real understanding, not memorization.

Q1. What problem does Linux solve, in your own words?

Notes: _____

Q2. What is the difference between normal behavior and a suspicious observation in this topic?

Notes: _____

Q3. Which command, log, packet, file, or response would you inspect first, and why?

Notes: _____

Q4. What could cause a misleading result or false positive here?

Notes: _____

Q5. How would you document your observation so another learner could reproduce it safely?

Notes: _____

Q6. Which security control reduces the risk associated with this topic?

Notes: _____

6. Revision Checklist

- I can explain linux fundamentals in my own words.
- I can identify the main terms without looking at the notes.
- I can perform the example in a controlled lab.
- I can interpret the output instead of only copying it.
- I know at least one defensive or troubleshooting use for the concept.
- I can document the result clearly and safely.

7. Safety & Scope

Use these commands and tools only on systems you own, a local virtual machine, a CTF, an intentionally vulnerable application, or a target for which you have explicit written permission.

Keep a written scope: target, allowed actions, time window, and stop conditions. Never scan or test random public systems.

When practicing, save your notes and outputs so you can explain what you observed rather than blindly running commands.

Next step: After Day-1, review your notes and connect this topic to the next subject in the series. The goal is not to memorize commands — it's to understand systems well enough to reason about security.

8. Quick Reference & Personal Notes

Topic: Day-1 — Linux Fundamentals

Three things I learned:

1. _____
1. _____
1. _____

Commands / concepts to memorize:

- `pwd` — show current directory
- `ls -la` — list files including hidden files
- `cd` — change directory
- `mkdir` — create directory
- `touch file` — create file
- `cp A B` — copy
- `mv A B` — move/rename
- `rm file` — remove file carefully
- `cat file` — display text
- `chmod` — change permissions

My lab result

What I tested: _____

What I observed: _____

What I would investigate next: _____

Final reminder: A strong cybersecurity learner can explain both the offensive concept and the defensive reason for learning it. Keep practice authorized, reproducible, and documented.